

SEDAR 19: South Atlantic and Gulf of Mexico Black Grouper and South Atlantic Red Grouper

Data Workshop

Life History Work Group

- The DW LH WG recognized the value of continuing the age workshops and exchange of otoliths in preparation of SEDAR data workshops. This will be especially important for species that have been recognized as relatively difficult to age.
- The DW LH WG also recognizes the value of similar workshops to discuss the interpretation of reproductive samples, and the possible exchange of histological sections between labs in preparation of SEDAR Data Workshops. This will be especially important for species that have been recognized as relatively difficult to stage.
- Since fecundity information is only available from the GOM and does not include estimates for ages less than 5 years, the DW LH WG recommends initiating a study to estimate fecundity and further identify spawning locations for all age classes in both the GOM and Atlantic populations.
- The data presented at the DW suggest a possible disjunct distribution in the Atlantic stock (NC-FL). The DW LH WG recommends a study to further investigate this by use of genetic, tagging, and other techniques.
- Improved collection and collection strategy for hard parts, in particular from the recreational sector.
- Increase of Fishery Independent data to include the entire area of red grouper distribution in the Atlantic.
- Virtually no information on the life history and distribution of juvenile red grouper (i.e. ages 0-2) is available. The DW LH WG recommends a study to gather information on these early stages.

Procedural recommendation:

- The DW recommends that the report of the natural mortality workshop organized by NMFS (Seattle, WA, August 2009) be made available to the DW LHW before the next SEDAR as a guide in the discussions concerning natural mortality.

Commercial Work Group

- Still need observer coverage for the snapper-grouper fishery
 - 5-10% allocated by strata within states
 - get maximum information from fish

- Expand TIP sampling to better cover all statistical strata
 - Predominantly by H&L gear
 - In that sense, we have decent coverage for lengths
- Trade off with lengths versus ages, need for more ages (i.e., hard parts)
- Workshop to resolve historical commercial landings for a suite of snapper-grouper species
 - Monroe County (SA-GoM division)
 - Historical species identification (mis-identification and unclassified)

Recreational Work Group

- Need more detailed information about where the fish are caught (depth, spatial, etc.)
- More detailed information on recreational discards, such as hooking location, depth fished, etc. that are likely to impact discard mortality and discard size/age.
- Additional information on sector (mode) differences.

Measures of Abundance Work Group

1. Expand fishery independent sampling to provide indices of abundance. The DW Panel noted that this recommendation has been the first on the list for virtually all previous SEDAR's in the south Atlantic.
2. Examine variability in catchability
 - Environmental effects
 - Changes over time associated with increases in technology and potential changes in fishing practices. This is of particular importance when considering fishery dependent indices.
 - Potential density-dependent changes in catchability. This is of particular importance for schooling fishes.
3. Conduct studies to examine how the behavior of fisherman changes over time and how these changes relate to factors such as gas prices and economic trends
4. Consider optimal sample allocation for species of interest when designing surveys to increase sample sizes.
5. Examine possible temporal changes in species assemblages. Such changes could influence how the Stephens and MacCall method is applied when determining effective effort.
6. Continue to expand fishery dependent at-sea-observer surveys. Such surveys collect discard information, which would provide for a more accurate index of abundance.

Assessment Workshop

1. Expanded fishery independent surveys of reef fishes in the Southeast, including red grouper, would greatly improve stock assessments.
2. More information on age/length composition of discards from various fleets would improve stock assessment of reef fishes in the Southeast, including red grouper.
3. More information on discard mortality rates would improve stock assessment of reef fishes in the Southeast, including red grouper.
4. The apparent stock separation of red grouper deserves further consideration. It may be desirable to develop appropriate spatial assessment models, if corresponding data requirements could be met. It may be desirable to research methods of spatial management (whether or not the assessment is spatially explicit).
5. More detailed spatial resolution of fishing effort would likely improve assessments.
6. Information on historical landings of reef fishes in the Southeast could lead to improved understanding of stock productivity and dynamics of stock assemblages.
7. Methods to characterize uncertainty in assessment results deserve further consideration. For avoiding overfishing, characterizing uncertainty is more than an academic exercise, particularly when relying on probabilistic methods to set catch levels.
8. Effects of new management measures (Amendment 16, seasonal closure) should be monitored.

Review Workshop

Members of the Data and Assessment workshops identified a number of shortcomings in the data available for red grouper, and the Review Panel (RP) felt that future research efforts should be focused on obtaining more precise estimates for parameters that displayed a strong potential effect and high uncertainty on the output of the assessment models. This opinion was reinforced by the fact that red grouper are not abundant, nor do they represent substantial fisheries; hence data acquisition efforts are hampered by both low abundances and low availability of samples from fishery sources. Many of the research recommendations have a reasonable biological basis, but a number are not directly linked to the assessment models used in the stock assessment. The RP felt that future research should focus on discard mortality, especially from the recreational fishery, acquiring better fishery-independent abundance estimates, improved methods for estimating catch by recreational anglers, improved age and growth data, and efforts to quantify linkages (i.e., recruitment effects) between western Caribbean and US stocks of red grouper. Given that fecundity data are not currently used in the stock assessment models, nor are histological gonadal stages utilized other than to distinguish mature from immature specimens, we suggest that these studies have lower priority than the research needs identified above. Studies directed towards identifying locations of spawning aggregations may be difficult to conduct for a species with low abundance, although such studies would be useful if spatially-based fisheries closures were to be employed for red grouper.

The RP recommends a strategic approach should be taken towards research for the snapper-grouper complex. The criteria which would be used to evaluate the strategy should be:

- Efficiency: for example sampling for sex ratio, length, and age could cover a range of species simultaneously.
- Impact: the resulting information should have clear implications for decision making.

To achieve this, managers and scientists will both need to be involved in developing a strategic research plan.

The RP recommends future research to determine which F metric behaves best under this management system.